

The Chinese University of Hong Kong, Shenzhen

Course Outline

1. Course identity

A. Course as listed at CUHK-Shenzhen

The information in this block should be exactly as approved by CUHK Senate. In case there are any differences, please explain in the table below.

Course code	MFE5190
Course title (English)	Credit Risk Modeling and Products
Course title (Chinese)	信用风险模型和产品
Units	3
Description (English)	The course introduces credit risk modeling and credit derivatives evaluation and management. It covers structural models of default risk, intensity-based modeling, risk structure of interest rates, credit default swaps, Collateralized Debt Obligations and related products, default-dependent structure modeling.
Description (Chinese)	本课程介绍信贷风险模型和信贷衍生工具的评估和管理。其主题涵盖了违约风险的结构性模型，强度为本模型，利率风险结构，信贷违约掉期，债务抵押债券及相关产品，和违约相关的结构性模型。

B. Corresponding course at CUHK

Please give details of the closest corresponding course at CUHK (as approved by CUHK Senate and listed in course list). If the course in Shenzhen maps to more than one course at CUHK, please make multiple copies of the block below.

Course code	
Course title (English)	
Course title (Chinese)	
Units	
Description (English)	
Description (Chinese)	

2. Course requisites

Please state prerequisites and co-requisites, in terms of courses at CUHK-Shenzhen or any other requirements.

A. Prerequisites

MFE5120 Investment Sciences or equivalent
MFE5130 Financial Derivatives or equivalent

B. Co-requisites

N/A

C. Others

N/A

3. Learning outcomes

Model the credit risk of major credit products
Conduct credit assessments using machine learning
Use credit derivatives to hedge credit risk
Implement credit regulatory standards: IFRS 9 and Basel III Framework Reforms

4. Course syllabus

Please highlight fundamental concepts involved in each topic to help students better understand what is covered in the course.

Credit theories

- Credit risk factors: EAD, LGD, PD, RM, NOB and CCC
- Credit risk measures: EL, 1-year EL and XCL
- Debt portfolio models

Credit derivatives

- Credit default swaps
- Credit linked notes
- Collateralized debt obligations
- Risk neutral valuation

Credit regulations

- The IFRS 9
- Basel III Framework Reforms
- Internal ratings based approach

Credit FinTech

- KNIME low code machine learning platform
- Generative AI

5. Assessment scheme

Component/ method	% weight
Term project	25
Mid-term test	25
Final examination	50

Remarks:

6. Grade descriptor

The grade descriptor should align with specific course learning outcomes, not a generic description applicable to all courses.

Type:**General**

Grade	Description
A	Demonstrate solid understanding of the course material in term project, mid-term test and final examination.
B	Demonstrate substantial understanding of the course material in term project, mid-term test and final examination.
C	Demonstrate fair understanding of the course material in term project, mid-term test and final examination.
D	Demonstrate limited understanding of the course material in term project, mid-term test and final examination.
E	Demonstrate unsatisfactory understanding of the course material in term project, mid-term test and final examination.

7. Feedback for evaluation

Please specify forms of evaluation to generate feedback on classes from students.

Term end course and teaching evaluations

Outside class feedbacks to lecturer and/or teaching assistant

Regular programme review

8. Readings

Textbook:

- 1.Credit Risk Modeling using Excel and VBA, 0470660929, Löffler and Posch, Wiley, 2ND, 2011, United States
- 2.Credit Risk Analytics: Measurement Techniques, Applications, and Examples in SAS, 1119143985, Baesen, Roesch, Scheule, Wiley, 1ST, 2016, United States
- 3.Managing Credit Risk Under The Basel III Framework, 171959970X, Lam, Kwan and Lai, CreateSpace Independent Publishing Platform, 3RD, 2018, United States

Others:

9. Course components

Activity	Hours/week
Lecture	6

10. Indicative teaching plan

Week	Content/ topic/ activity
1	Credit risk factors, Credit products, Credit ratings and FICO scores
2	Benchmark model, Credit machine learning
3	Credit portfolio models, Credit risk monitoring, Credit risk controls
4	Credit default swaps, Multiple name credit derivatives, Synthetic credit derivatives
5	Financial tsunami 2008, Credit provision and capital charge
6	Internal ratings based approach, Supervisory credit exposures, Generative AI for credit regulations
7	Revision, Final examination

11. Implementation plan

The implementation plan may vary from year to year. Please indicate expected enrolment, and number of sections. Example: 150 students for lecture (x 2); 30 students for tutorials (x 10).

60 students for lecture

12. Academic honesty and plagiarism

A course outline may include subject-specific requirements on plagiarism, in addition to the University established policies.

Students must complete all coursework and assessments independently, unless AI tool use is explicitly permitted. Unauthorized or improper use of AI may constitute academic dishonesty and harm learning. Misuse includes submitting AI-generated work as one's own, using AI to cheat, or employing AI unethically. Always obtain permission before using AI in academic work, and be sure to acknowledge any AI use in submissions.

13. Approval

Has the course title been included in the programme submission approved by CUHK Senate? Are there any differences? Have the details (as in this document) been approved at School or other level in CUHK- Shenzhen?

Submit application

14. Any other information

15. Version date

Version number	1
As of (date)	2026-04-17